

COS 202 PRACTICAL ASSIGNMENT REPORT

Title:

Development of a Mathematical Calculator and Personal Pocket CGPA Calculator Using Python

Name:

Adepoju Betty pamilerin

Matric Number: 241201115

Department: computer science

Course Code:

COS 202

INTRODUCTION

Python is a high-Level programming language that is widely used for developing software applications because of its simplicity and flexibility. This assignment involves the development of two Python programs: a Mathematical Calculator and a Personal Pocket CGPA Calculator. These programs demonstrate the use of Python programming concepts such as variables, arithmetic operators, loops, user input, and selection control statements.

QUESTION 2: MATHEMATICAL CALCULATOR

Objective

To develop a simple mathematical calculator that performs basic arithmetic operations.

Program Description

The Mathematical Calculator allows the user to perform arithmetic operations such as addition, subtraction, multiplication, division, floor division, power, and modulus. The user selects an operation from the menu, enters two numbers, and the program displays the result. The calculator continues to run until the user selects the OFF option.

QUESTION 3: PERSONAL POCKET CGPA CALCULATOR

Objective

To develop a Personal Pocket CGPA Calculator that calculates a student's CGPA and determines the appropriate class of degree.

Program Description

The Personal Pocket CGPA Calculator accepts the student's total grade points and total credit units as input. It calculates the CGPA and uses Python selection statements (if, elif, and else) to determine the student's class of degree based on the calculated CGPA.

CONCLUSION

The two Python programs were successfully developed and tested. The Mathematical Calculator performs the required arithmetic operations correctly, while the Personal Pocket CGPA Calculator accurately computes a student's CGPA and determines the corresponding class of degree. The assignment demonstrates the practical application of Python programming concepts in solving mathematical and academic problems.